

Design & Development of a 5-Band Audio Equalizer

Report submitted by

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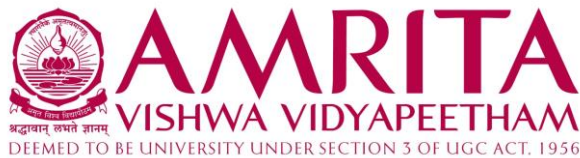
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Table of Contents

List of Tables	i
List of Figures	ii
1. Motivation	1
2.Problem Statement	1
2.1 Design Specifications	1
3. Design Procedure	2
3.1 Step 1 – Audio Input	2
3.2 Step 2 – Frequency Band Separation	2
3.3 Step 3 – Gain Control	3
3.4 Step 4 – Summing Stage	3
3.5 Step 5 – Audio Output	3
3.6 Components & Budget	4
4. Circuit Diagram/ Board Layout/ Schematic	5
5. Simulation Results/ Discussion	5
6. Implementation/ Prototyping Results	8
7. Discussion & Conclusion	9
8. References	9

List of Tables

Table No.	Table Caption	Page No.
Table 2.1	Target Frequency Bands of equalizer	1
Table 3.1	Design Values of Resistors, Capacitors	3
Table 3.2	Components used and Budget	4

List of Figures

Figure No.	Figure Caption	Page No.
Fig. 3.1	MFB BPF Illustration	2
Fig. 3.2	Circuit Diagram	4
Fig. 4.1	Working PCB Model	5
Fig. 4.2	PCB Layout	5
Fig. 5.1	Input Signal	5
Fig. 5.2	1 st , 2 nd Stage Time Domain Outputs	5
Fig 5.3	1 st , 2 nd Stage Freq. Domain Outputs	6
Fig 5.4	Final Output Time domain	6
Fig 5.5	1, 2, 3 Bands Time Domain Outputs	6
Fig 5.6	Band 1 Output Frequency Domain	6
Fig 5.7	Band 2 Output Frequency Domain	6
Fig 5.8	Band 3 Output Frequency Domain	7
Fig 5.9	Band 4 Output Frequency Domain	7
Fig 5.10	Band 5 Output Frequency Domain	7
Fig 5.11	Overall Output Frequency Domain	7
Fig 6.1	Input Audio FFT	8
Fig 6.2	Output Audio FFT (Band 2 Boosted)	9

1. MOTIVATION

Audio signals consist of a wide range of frequencies that contribute differently to the overall sound quality. Controlling these frequency components is important in many audio systems for improving clarity and tonal balance. Audio equalizers enable adjustment of specific frequency bands, allowing certain components of the signal to be amplified or attenuated. Designing a 5-band analog audio equalizer using TL064 operational amplifiers involves the application of active filter design and analog signal processing concepts. Implementing such a circuit provides practical understanding of band-pass filters, frequency band separation, and gain control across different parts of the audio spectrum.

2. PROBLEM STATEMENT

The goal of the project is to design and implement a 5-Band Analog Audio Equalizer capable of independently controlling different frequency components of an input audio signal.

2.1 Design Specifications:

- Design a 5-band analog audio equalizer using Multiple Feedback (MFB) band-pass filters.
- Implement five frequency bands covering the major audio spectrum.
- Provide independent gain control for each band using potentiometers.
- Combine outputs of all bands using a summing amplifier.
- Amplify the final signal to drive wired earphones.

Table 2.1: Target frequency bands of equalizer:

Band	Frequency Range
Band 1	60 – 300 Hz
Band 2	300 Hz – 1 kHz
Band 3	1 – 4 kHz
Band 4	4 – 10 kHz
Band 5	10 – 16 kHz

3. DESIGN PROCEDURE

The design of the 5-band audio equalizer is based on active analog filters implemented using TL064 operational amplifiers. Each band isolates a specific range of frequencies and allows the user to adjust its amplitude using a potentiometer.

3.1 Step 1 – Audio Input

The audio signal is obtained from a wired earphone jack, which converts the audio waveform into an electrical signal. This signal serves as the input to the equalizer circuit.

3.2 Step 2 – Frequency Band Separation

The frequency spectrum is divided into five bands using Multiple Feedback (MFB) band-pass filters. Each band is implemented using two cascaded second-order filters, resulting in a fourth-order band-pass filter. This configuration improves frequency selectivity and reduces interference from neighbouring bands.

The center frequency of each filter is determined using

$$f_c = \frac{1}{2\pi\sqrt{R_a R_b C_1 C_2}}$$

Where R_a, R_b are resistors in the feedback network
 C_1, C_2 are capacitors of the filter.

Fig 3.1: MFB BPF Illustration

MULTIPLE FEEDBACK BAND-PASS DESIGN EQUATIONS

$$\frac{-H \omega_0 s}{s^2 + \alpha \omega_0 s + \omega_0^2} \quad (1)$$

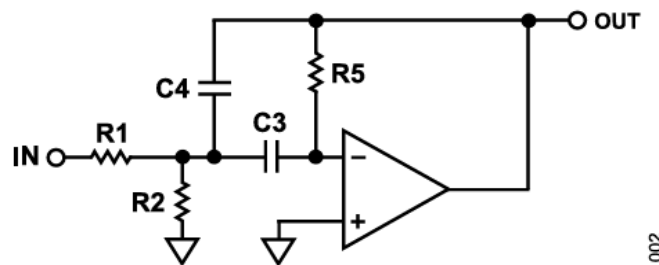


Figure 2.

$$\frac{V_O}{V_{IN}} = \frac{-s \frac{1}{R_1 C_4}}{s^2 + s \frac{(C_3 + C_4)}{C_3 C_4 R_5} + \frac{1}{R_5 C_3 C_4} \left(\frac{1}{R_1} + \frac{1}{R_2} \right)} \quad (2)$$

Derivation of Resistor Values for 60–300 Hz Band:

Band edges: $f_L = 60$ Hz, $f_H = 300$ Hz

Fixing the capacitor values:

$C_1 = 6.8$ nF (left biquad), $C_2 = 68$ nF (right biquad).

Centre frequency and quality factor are given by:

$$f_0 = \frac{1}{2\pi C \sqrt{R_a R_b}}, \quad Q = \frac{1}{2} \sqrt{\frac{R_b}{R_a}}.$$

$$R_a = \frac{1}{4\pi C Q f_0}, \quad R_b = 4Q^2 R_a.$$

Where R_a and R_b are the input and feedback resistors respectively

partitioning the band such that the two centres are symmetric around the overall geometric mean:

$$f_{gm} = \sqrt{60 \cdot 300} = 134.164 \text{ Hz}$$

Two centres f_{02} and f_{01} with $f_{01} \cdot f_{02} = f_{gm}^2$ and chosen spread to produce smooth overlap.

$$f_{02} \approx 88.14885 \text{ Hz (lower biquad)}$$

$$f_{01} \approx 254.46382 \text{ Hz (upper biquad)}$$

For the left (upper-center) biquad with $C_1 = 6.8$ nF:

- chosen center $f_{01} = 254.463817$ Hz
- chosen $Q_1 = 0.9784921096$

For the right (lower-center) biquad with $C_2 = 68$ nF:

- chosen center $f_{02} = 88.148852$ Hz
- chosen $Q_2 = 0.8850612032$

3.3 Step 3 – Gain Control

After filtering, the signal of each band passes through a potentiometer-based amplifier stage. By adjusting the potentiometer, the user can increase or decrease the gain of that particular frequency band

3.4 Step 4 – Summing Stage

The outputs of all five filter stages are combined using an operational amplifier summing circuit implemented with the TL064. This stage adds the weighted contributions of each band to produce the final equalized signal.

3.5 Step 5 – Audio Output

The summed output signal is provided to wired earphones, allowing the listener to hear the effect of frequency adjustments across the five bands.

Table 3.1: Design Values of Resistors and Capacitors

Frequency Band	R1	R2	R3	R4	R5	R6	C1	C2
60–300 Hz	47K	68K	180K	15K	15K	47K	6.8nF	68nF
300–1000 Hz	1.5K	1K	47K	3.3K	2.2K	10K	100nF	100nF
1–4 kHz	1K	680Ω	3.3K	1K	680Ω	2.2K	33nF	100nF
4–10 kHz	1K	470Ω	3.3K	1K	470Ω	3.3K	15nF	33nF
10–16 kHz	33K	1.5K	100K	33K	2.2K	100K	1nF	1nF

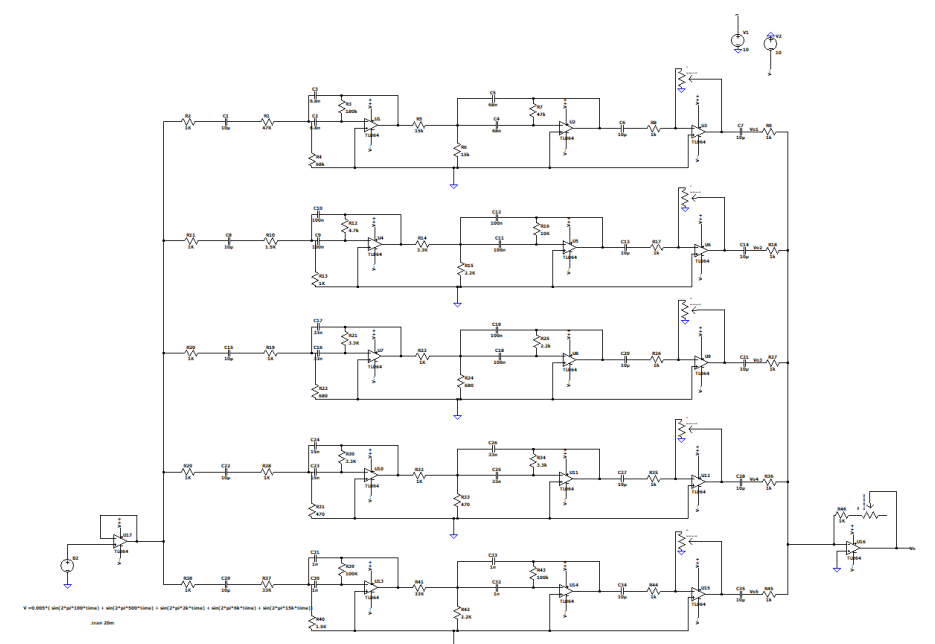
3.6 Components & Budget

In the budget section, the cost of your project will be detailed, with a break up for the boards, components and services (like PCB) required. The costing should be on actuals and should not be inflated or arbitrary.

Table 3.2: Components used and Budget

S. No	Component	Specification	Quantity	Cost
1	OpAmp IC's	TL064	5	100
2	Potentiometer	10K Ω	6	120
3	Aux Cable	3.5mm	1	120
4	Audio Jack	SJ2-3574A-SMT-TR	2	300
5	Resistors	680 Ω , 470 Ω , 1K Ω , 1.5 K Ω , 2.2K Ω , 3.3K Ω , 10 K Ω , 15 K Ω , 33 K Ω , 47K Ω , 68 K Ω , 100 K Ω , 180 K Ω	33 (2, 2, 5+2, 2, 2, 5+1, 1, 2, 2, 3, 1, 2, 1,)	50
6	Capacitors	1nF, 6.8nF, 15nF, 33nF, 68nF, 100nF	2, 1, 1, 2, 1, 3	30
7	Printed PCB	2 layer	5	1398
8	Connectors		4	
9	IC holders	Standard 7 pin	5	15

Fig 3.2: Circuit Diagram



4. Circuit Diagram / Board Layout / Schematic:

Fig 4.1: Working PCB Model

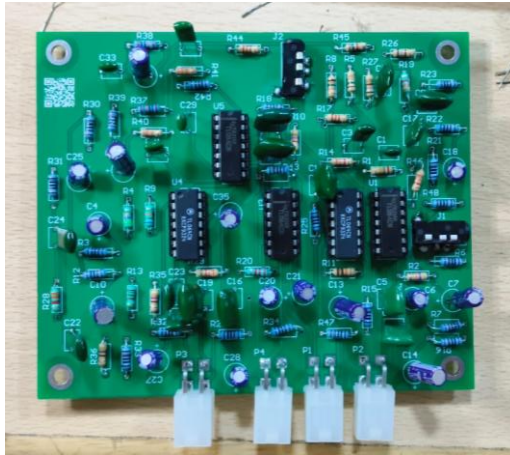
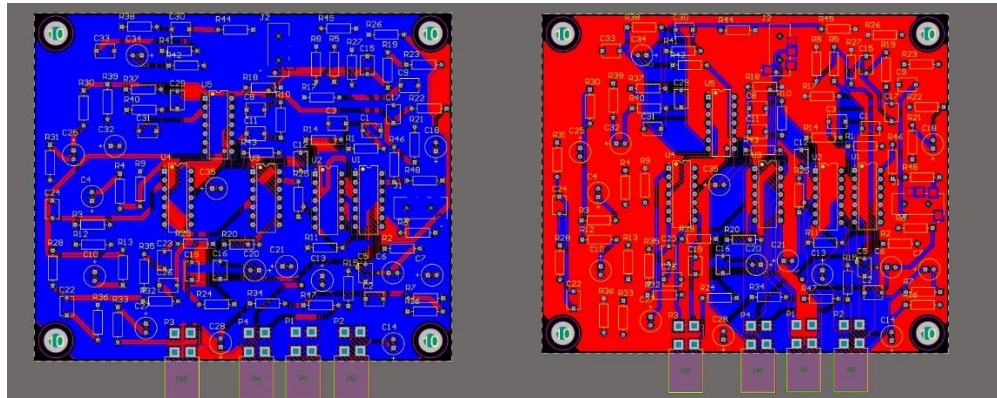


Fig. 4.2: PCB layout



5. Simulation Results / Discussion:

Fig 5.1: Input source – Sum of 5 signals (5mV sine waves) in each band (100Hz, 500Hz, 2KHz, 6KHz, 15KHz)

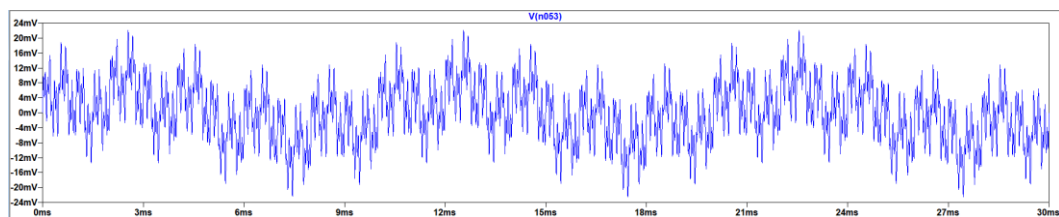


Fig 5.2: 1st stage, 2nd stage bandpass outputs (time domain)

(Stronger rejection of other frequencies as order increases)

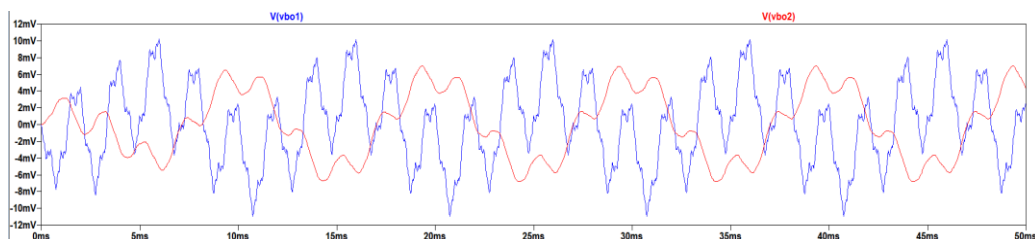


Fig 5.3: 1st stage, 2nd stage bandpass outputs (frequency domain)

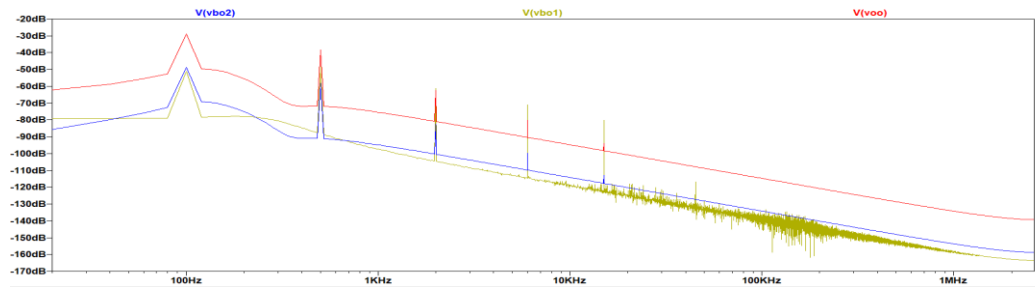


Fig 5.4: Final Output of Audio equalizer: (after merging outputs of all bands)

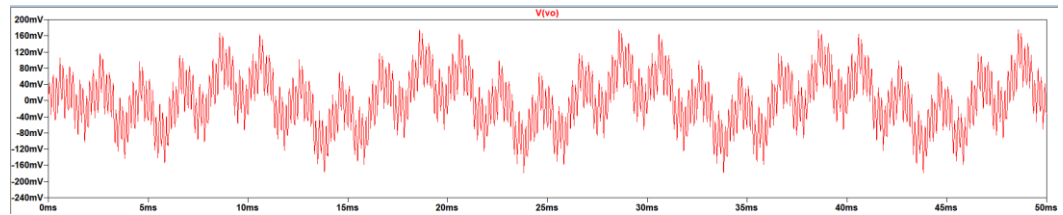


Fig 5.5: Time domain representation of 1,2,3 bands final output

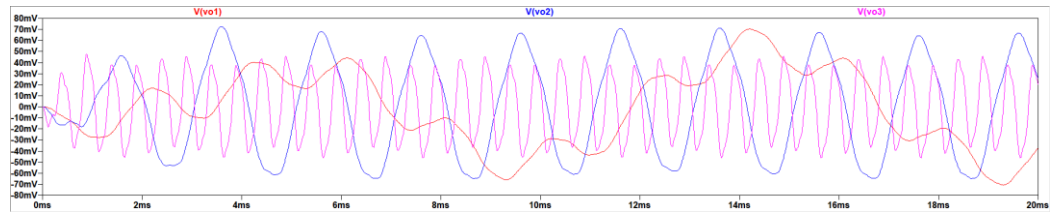


Fig 5.6: Frequency domain representation band 1 output

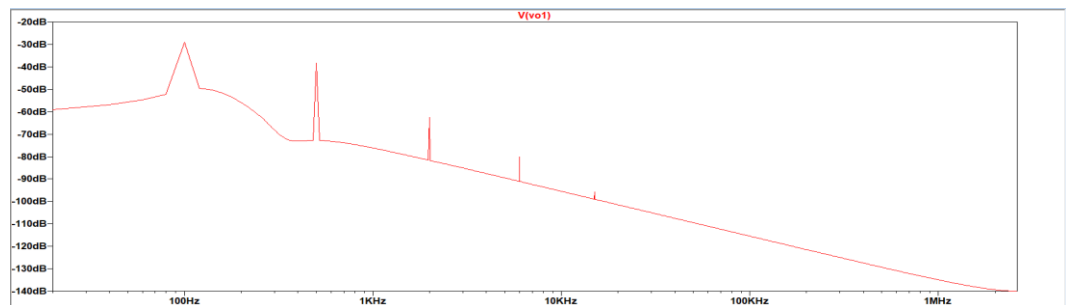


Fig 5.7: Frequency domain representation band 2 output

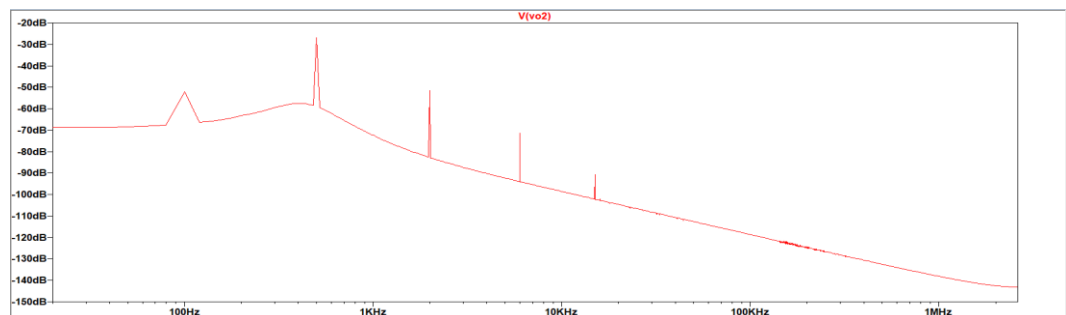


Fig 5.8: Frequency domain representation band 3 output

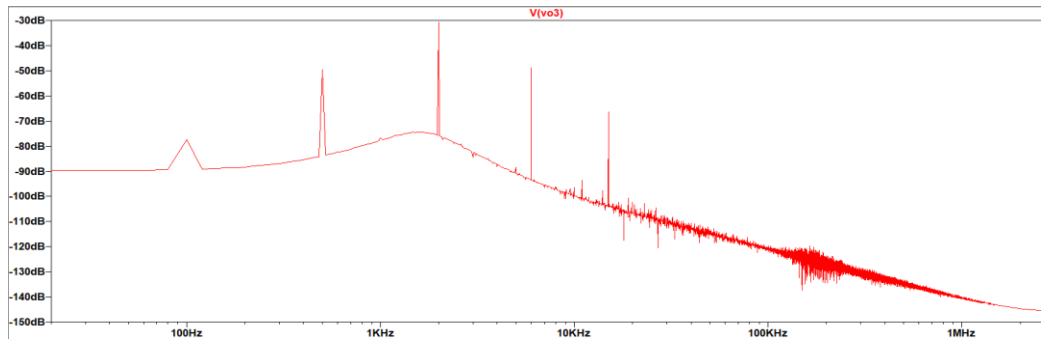


Fig 5.9: Frequency domain representation band 4 output

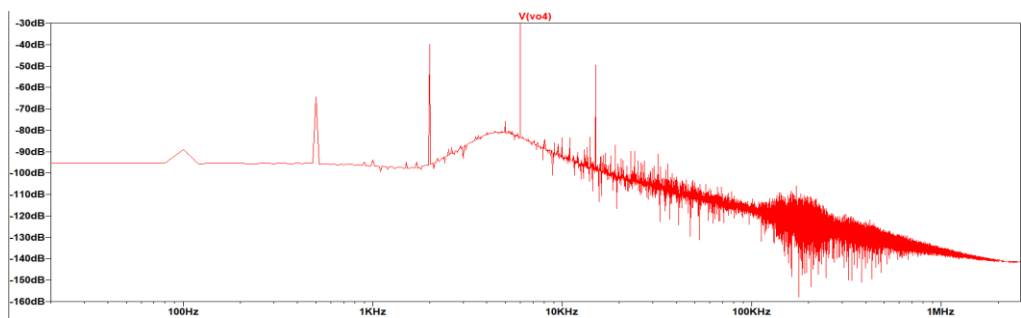


Fig 5.10: Frequency domain representation band 5 output

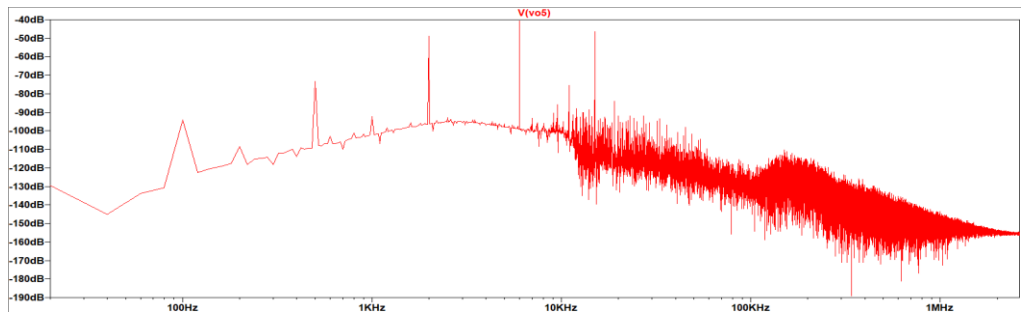
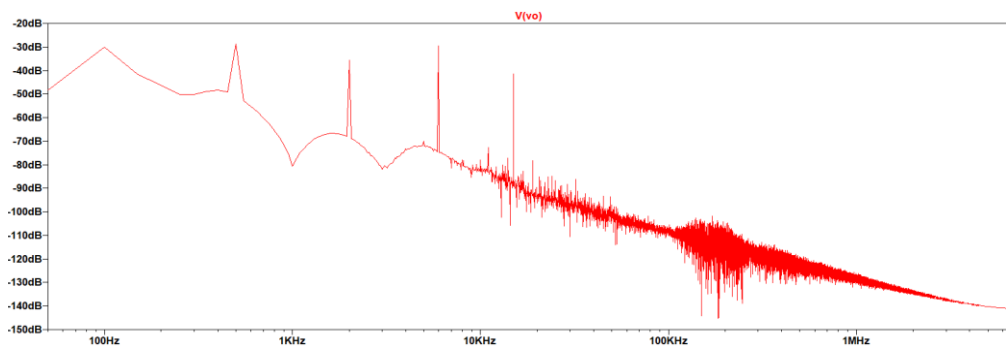


Fig 5.11: Frequency domain representation Final Output



Analysis: The simulation results verify the correct operation of the designed 5-band audio equalizer. The time-domain outputs show that the cascaded band-pass filter stages progressively isolate the desired frequency components while attenuating others. Frequency-domain plots confirm the band-pass behaviour of each filter with clear peaks in their respective ranges. The combined output represents the weighted contribution of all five bands after gain adjustment, demonstrating effective frequency selectivity and independent band control. Minor disturbances near the higher frequency bands could be reduced by using higher-order filters or improving band-pass selectivity.

6. Implementation / Prototyping results

Fig 6.1: FFT - Input Audio Signal

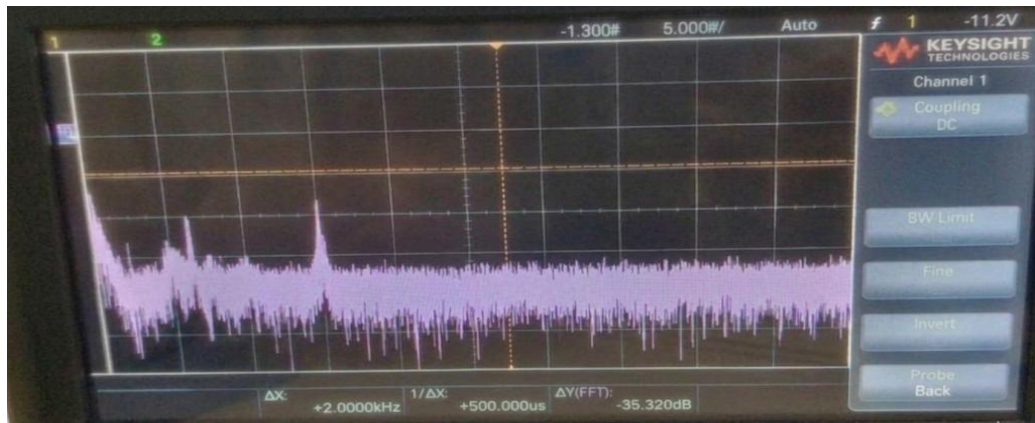
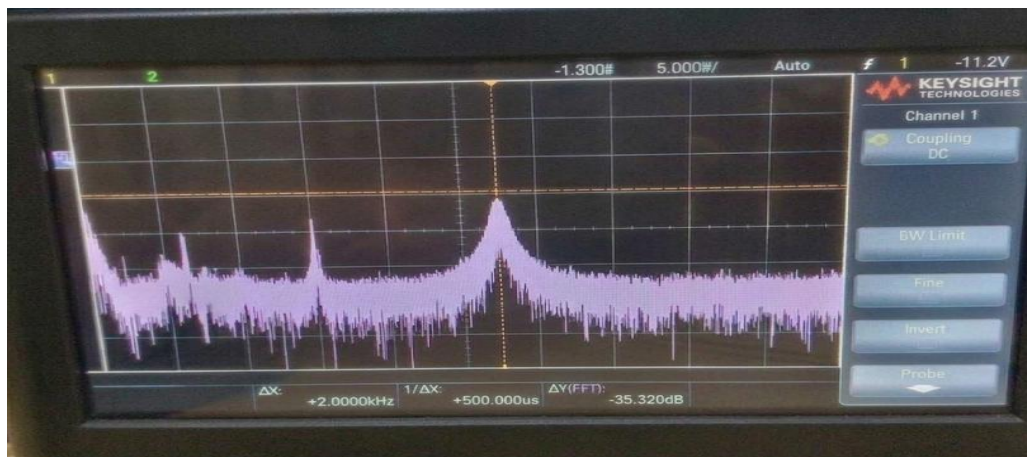


Fig 6.2: FFT- Output Audio Signal - 2nd Band Boosted



Analysis: The hardware implementation of the equalizer using TL064 operational amplifiers successfully demonstrated the practical behaviour of the designed circuit. The FFT of the input audio signal showed multiple frequency components present in the original audio.

After passing through the equalizer, adjustments made to the potentiometers modified the amplitude of the corresponding frequency bands. For example, boosting Band 2 increased the spectral magnitude in the mid-frequency region in the output FFT. The experimental results closely follow the behaviour observed in simulation, confirming that the circuit effectively performs frequency band separation and adjustable gain control for audio signals. Minor variations in response were observed due to component tolerances and practical circuit non-idealities. A slight disturbance in higher frequency audio components was also noticed, which may arise from noise and component tolerances in practical circuits.

7. Discussion & Conclusions:

The designed 5-band audio equalizer successfully demonstrated the use of Multiple Feedback (MFB) band-pass filters implemented using TL064 operational amplifiers to separate and control different audio frequency bands. Simulation results showed clear band-pass behaviour for each stage, and the combined output confirmed that the amplitude of different frequency components can be adjusted independently. The hardware implementation produced results similar to the simulation, where adjusting the potentiometers modified the corresponding frequency bands in the output audio.

Slight disturbances were noticed at higher frequencies, which may be reduced by using higher order filters or improved circuit layout. Further improvements such as implementing the design on a PCB and using precision components can enhance stability and reduce noise.

8. References

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